Glossary

Absolute path: the location of a file specified from the root directory (the full path).

Abstraction: having a higher-level, simplified model to represent a complex system. It allows you to focus on the core ideas or concepts that matter, without being overly concerned about the intricate details of implementation.

Access modifiers: the mechanisms provided by the programming language to control visibility of methods and variables within an object.

Accessor: a public method that allows external code to "access" the value of a private instance variable within an object; also known as "getter method" as it "gets" the value.

Activation function: a mathematical function applied to the output of a neuron that is used to determine whether or not the neuron should be activated (considered to be "on").

Aggregate functions: functions used to perform calculations on multiple records based on a given field, e.g. AVERAGE, COUNT, MAX, MIN, SUM.

Aggregation: where one object "has" another object as part of it, but the two objects can exist independently of each other.

Algorithm: a finite sequence of instructions that needs to be followed step-by-step to solve a problem.

Amplitude: the magnitude of change in a sound wave, representing the loudness or intensity of the sound.

Analogue: a continuous signal that represents varying physical quantities, such as sound waves, which varies smoothly over a range; digital represents data in discrete binary values (0s and 1s), enabling precise and error-resistant processing.

Arithmetic operator: a character that is used to perform a calculation.

Artificial intelligence: computer technology able to perform tasks and make decisions in a manner that imitates human intelligence. There are two main forms of AI: narrow (or weak) AI is designed to perform specific tasks or solve specific types of problems; general (or strong) AI processes human-level intelligence and can operate across a range of domains. While speculation persists that general AI is "close", at this time only narrow AI technology is available.

ASCII (American Standard Code for Information Interchange): a character-encoding standard used to represent text in computers and other devices, defining a numerical value for each symbol and character commonly used in the English language.

Assignment: to set, reset or copy a value into a variable.

Association rule: a process of finding patterns of co-occurrence in data; this means, given the presence of one item in a record, how likely it is that another item will be present.

Atomic: each attribute in a table containing indivisible values (values that cannot be broken down into more detailed sub-values).

Attribute: a data item or a characteristic of an entity; a column in a table.

Backpropagation: backpropagation of errors is the most commonly used technique for training artificial neural networks. The gradient of the loss function is calculated, and used to update parameters such as weights, in the opposite direction of the gradient to reduce the overall error.

Base case: a terminating solution (that is not recursive) to a process.

Basic Multilingual Plane (BMP): the most commonly used characters and symbols for almost all modern languages.

Bidirectional bus: a bus that can transfer data in both directions.

Big O notation: used to find the upper bound (worst-case scenario or the highest possible amount) of the growth of a function; the longest time or space required to turn the input into output.

Binary operator: an operator that requires two operands (values).

Binary search: a method of searching an ordered array (list) by repeatedly checking the value of the middle element and disregarding the half of the data structure that does not contain the searched element.

Bit: binary digit; a single digit, either 1 or 0.

Bitmap: a type of digital image composed of a grid of pixels, each holding a specific colour value, representing the image in a rasterized format.

Boolean: a data type to represent one of the two possible values: true or false.

Boolean operator: a character that represents a specific logical operation that is used to produce a true or false outcome.

Breakpoint: a marker to interrupt the execution of code for debugging purposes.

Brute force: a method of breaking a cipher by systematically trying every possible key until the correct one is found.

Bubble sort: a sorting algorithm that compares adjacent values and swaps them if they are in an incorrect order.

Buffering: the process of temporarily storing data in a memory area (buffer) while it is being transferred between two devices or processes, helping to manage differences in data-flow rates and ensuring smooth, uninterrupted operation.

Business intelligence: technologies, applications and practices for collecting, integrating, analysing and presenting business information.

Byte: 8 bits.

Cache hit: when the CPU requests data and it is found in the cache memory.

Cache miss: when the CPU requests data and it is not found in the cache memory, necessitating retrieval from slower main memory or storage.

Caching: the process of temporarily storing frequently accessed data in a high-speed storage area (cache) to reduce access time and improve system performance by enabling quicker retrieval of the data.

Cardinality: the maximum number of times an instance in one entity can be associated with instances in the related entity.

Char: a data type used to represent one single character, digit or symbol.

Child: any node that has a direct link from a parent node positioned above it, potentially having further child nodes of its own.

Classification: machine learning where the output generated should be a category, chosen from among a discrete set of categories available.

Classification techniques: where a machine learning model has been trained to identify, from a predefined list of categories, which category (or class) the input data would most likely be associated with.

Cloud database: a database that runs on cloud computing platforms, providing scalability, high availability and flexible resource management.

Clustering techniques: where data is grouped into clusters based on similarity or proximity to each other without any labels provided to help indicate the correctness of associating any individual datapoint to the cluster assigned.

Colour depth: also known as "bit depth"; the number of bits used to represent the colour of each pixel in a digital image, determining the range and precision of colours that can be displayed.

Comment: a note that explains some code, which will be ignored at compilation stage.

Composite key: a set of attributes that form a primary key.

Composition: where objects are composed of other objects, forming a "has-a" style of relationship. The objects that comprise the internal objects cannot exist independently of the containing object.

Computational thinking: a toolkit of available techniques for problem-solving; its fundamental concepts are abstraction, decomposition, algorithmic thinking and pattern recognition.

Computer network: a system that connects computers and other devices to share resources (digital or physical) and information.

Concatenation: joining strings together.

Conceptual schema: an abstract model describing the structure of the data without considering how it will physically be implemented.

Confusion matrix: a simple pictorial means of representing how well a machine learning model is performing.

Constructor: a special method within a class that is automatically executed during instantiation; its main task is to initialize any instance variables required before an instance of the object can be used by other code.

Convolution: a mathematical operation that combines two functions to produce a third function. In the context of a convolutional neural network being used for image processing, convolution applies filtering functions to the pixels in an input image to compute distinctive features from the data.

Curse of dimensionality: each feature in a machine learning model adds another dimension to the overall model the algorithm is attempting to map and create generalizations about; the curse of dimensionality refers to the problem that occurs when there are too many dimensions relative to the quantity of data available, so that patterns cannot be meaningfully observed.

Data definition language: language that is used to create, modify and remove data structures from a relational database.

Data manipulation language: language that is used to add, modify, delete and retrieve data stored in relational databases.

Data mining: the process of sorting through large data sets to identify patterns and relationships that can help solve business problems through data analysis.

Data sparsity: how "spread out" data points are from each other in a model.

Data storage: storage of data within primary or secondary memory.

Data type: defines the type of value a variable or data structure has and what type of mathematical, relational or logical operations can be applied without causing an error.

Data warehouse: a specialized type of database designed for analytical purposes rather than transactional processing.

Database: an organized collection of structured information or data that can be accessed in different ways.

Database schema: an architecture showing how data is organized and how the relationship between data is managed.

Debugging: finding and fixing errors in a program.

Debugging tools: software applications or utilities used by developers to identify, analyse and fix bugs or issues within a program by inspecting code, variables and execution flow.

Decision tree: a graphical representation of conditions that will result in a classification decision being made; think of it as a decision-making flowchart that the machine learning model creates.

Declaration: a language construct specifying the properties of an identifier.

Decomposition: breaking down complex problems into smaller, more manageable parts.

Decrement: to decrease a value by another value (usually by one).

Deep learning: a subset of machine learning that uses an artificial neural network to imitate the design of the human brain to find generalizations in complex data that can be used for decision-making.

Defragmentation: the process of reorganizing the data on a hard drive so that files are stored in contiguous blocks, reducing fragmentation and improving access speed and overall system performance.

Denormalization: deliberately allowing for data redundancy in a database design to improve the performance of queries.

Dequeue: a method of deleting an element from the front of a queue.

Device drivers: specialized software programs that allow the operating system to communicate with and control hardware devices, e.g. printers, graphics cards or network adapters, by providing the necessary instructions and protocols.

Direct access: a method of access where elements are directly retrieved by using their index (position).

Distributed database: a database made of two or more files located on different sites on the same network or on completely different networks.

Domain name: a human-readable name assigned to a specific IP address on the internet, e.g. www.example.com.

Double: a data type used to represent a decimal number.

Dynamic data structure: a data structure that can grow or decrease at runtime, with elements stored in memory locations that are chained together, but not necessarily contiguous.

Encapsulation: bundles data and the methods that manipulate that data together into a single object. It serves to hide the implementation details of the object from outside code.

Encryption: the conversion of information or data into a mathematically secure format that cannot be easily understood by unauthorized people.

Encryption key: a string of characters or numbers used by an encryption algorithm to encode or decode data. It is the values that are input into the mathematical functions responsible for scrambling or descrambling the data.

Enqueue: a method of inserting an element at the rear of a queue.

Entity: a living or non-living thing that can have data stored about it that can be described, e.g. a person, a chair or an aeroplane.

Entity-relationship diagram: a visual representation of the entities in a database and the relationship between them.

Exception: an unexpected event that stops the execution of a program, e.g. division by 0.

Exception handling: a process of responding to an exception, so the system does not halt unexpectedly.

Extract: to gather data from various operational databases, flat files, APIs, etc.

Factory pattern: a design pattern that provides an alternative interface for creating objects in contrast to normal constructor-based instantiation.

Feature: a numeric property that can be used to contribute a data point for a machine learning algorithm to train on. Think of it as a variable in your data set.

File extension: a suffix at the end of a filename that indicates the file type and the program associated with opening or processing that file (e.g. .docx for Word documents, .jpg for images).

Firewall: a security system (hardware or software) that monitors and controls incoming and outgoing network traffic based on a set of security rules.

First In First Out principle: when the first element inserted is the first element removed.

First normal form: the status of a relational database in which entities do not contain repeating groups of attributes.

Float: a data type used to represent a decimal number.

Floating-point division: division in which the fractional part is kept.

Foreign key: an attribute in a table that refers to the primary key in another table.

Frame: a single image in a sequence of images that makes up a video or animation.

Full functional dependency: where dependent attributes are determined by the determinant attributes.

Function: a set of statements that can be grouped together and called in a program as needed; they always return at least one value.

Functional dependency: a relationship that exists between attributes, where one set of attributes (the determinant) determines the value of the other set (the dependent).

Functional testing: testing concerned with the behaviour of the application – specifically, whether it meets the requirements specified. This type of testing evaluates the software by providing inputs and examining the outputs, without considering how internal systems work. You are testing each of the success criteria on the whole application from the user's point of view.

Gateway: a device that connects different networks together and manages the traffic flow between them; often used to connect a local network to the internet.

General case: a process where the recursive call takes place.

Generative AI: a form of artificial intelligence capable of generating text, images, audio, video and other digital artefacts, usually in response to a prompt. It is a form experiencing rapid advances at the time of writing.

Genetic algorithm: imitates the concept of survival of the fittest and evolution by testing a population of possible solutions to a problem, using properties from the best-performing solutions to create a new population of possible solutions, and then repeating the process until a suitably performing solution has been identified.

Global variable: a variable that exists throughout a program.

Hash table chaining: a collision-resolution technique in hash tables where each bucket or index in the array can store multiple elements in the form of a linked list, allowing more than one entry to be stored at the same index.

Hashing algorithm: a function that converts input data of any size into a fixed-size string of characters, which typically represents the data in a compressed and seemingly random format and is used primarily for indexing and retrieving items in databases more efficiently.

Heap space: a region of dynamically allocated memory managed by the operating system where programs store variables and data structures that require memory allocation during runtime, allowing for flexible memory usage that can grow and shrink as needed by the application.

High load factors (hash tables): a condition where a sizeable portion of the hash table's slots are filled, leading to increased collisions and potentially degraded performance, due to more frequent need for collision resolution mechanisms.

High-frequency data: correspond to rapid changes in pixel values, representing fine details, edges and textures.

Hyperparameter: a parameter (or value assigned to a variable) that is set before the learning process, which guides the algorithm as it learns.

Hypervisor: software that creates and manages virtual machines by allowing multiple operating systems to run simultaneously on a single physical machine, sharing the underlying hardware resources.

Identifier: a lexical token that names the language's entities.

Image resolution: the number of pixels contained within a digital image, typically expressed as the dimensions (width × height) in pixels, and sometimes as the pixel density (PPI / DPI) for print quality.

Increment: to increase a value by another value (usually by one).

Inheritance: where a class takes a copy of an existing class as the starting point for all its internal methods and variables. These can then be overridden and extended upon to provide additional functionality, as required.

Initialization: assigning an initial value to a data structure.

In-memory database: a database that stores data entirely in the main memory (RAM) rather than on disk, providing extremely fast read and write operations.

Instantiation: the line of code that declares a new object variable based on the template code provided by a class, which then executes the constructor to initialize the object.

Integer: a data type used to represent a whole number.

Integer division: division in which the fractional part is discarded.

Interface: a contract that specifies a set of methods a class must implement, without defining how these methods are implemented, serving as a blueprint that promotes modularity, flexibility and abstraction in software development. This structure allows different classes to implement the same interface in diverse ways, while ensuring they provide the functionalities declared by the interface.

Internet: a global network of computer networks that are interconnected with each other and communicate through standardized protocols.

Interrupt: a signal sent from a device or software to request the processor's attention; the processor will stop its current activity until the interrupt has been serviced.

Interrupt handling: handling interrupt requests.

Interrupt service routine (ISR): a special function in a computer system that automatically executes in response to an interrupt signal, handling specific tasks, e.g. processing input from hardware devices or managing system events, before returning control to the main program.

IP address: a set of numbers that uniquely identifies each computer based on the Internet Protocol (either version 4 or version 6).

kHz (kilohertz): a unit of frequency equal to 1000 cycles per second, commonly used to measure the sampling rate of audio signals.

K-nearest neighbours: where data points are categorized based on the categories of the nearest points around them in the data set; *k* is a variable representing how many of those nearest points should be used to "vote" and determine what category to assign the new value.

Last In First Out or First In Last Out principle: the last element inserted is the first element removed.

Latency: the delay between the initiation of an action and the corresponding response, often referring to the time it takes for data to travel from its source to its destination in a network or system.

Leaf: a node that does not have any children, representing the endpoints of a binary search tree's branches.

Least significant bit (LSB): the rightmost bit in a binary number, representing the smallest value position (20 or 1).

Linear regression: a machine learning algorithm that seeks a linear line of best fit for a given data set, from which extrapolations can be made.

Linear search: a method of searching, in which each element is checked in sequential order.

Load: to load transformed data into a data warehouse.

Load balancing: the process of distributing network or application traffic across multiple servers or resources to ensure optimal performance, reliability and availability, preventing any single server from becoming overwhelmed.

Local area network: a system that connects computers and other devices within a small geographical area, such as an office or home.

Local variable: a variable that exists only within the block of code where it is defined.

Logic error: an error in a program that makes it operate incorrectly; it will not crash the program.

Logical schema: a detailed design of the structure of tables (fields and data types), relationships between tables and constraints.

Loop / iteration: a repetition.

Low-frequency data: correspond to slow changes in pixel values, such as broad areas.

Machine learning: a branch of AI where computers learn from data and experiences to perform specific tasks or solve specific problems, without being explicitly programmed to do so.

Maintainable code: clear, easy-to-read and modify code that can be reused within the same program or in other programs, by the same or other programmers.

Malware: a general term for any software designed with malicious intent, e.g. viruses, worms, trojans, spyware and ransomware, which can damage systems, steal data or disrupt operations.

Matrix and vector multiplications: fundamental operations in machine learning and graphics that involve complex mathematical calculations.

Memory dump: a process where the contents of a computer's memory are captured and saved, typically for the purpose of diagnosing and debugging software issues.

Metadata: information that describes other data, providing context and details about the data's content, structure and attributes. In the context of digital images, metadata includes such information as the image's dimensions, colour depth, creation date, author, camera settings and other properties that help with managing, understanding and using the image effectively.

Middleware: software that connects different applications, allowing them to communicate and share data. It helps different parts of a computer system work together smoothly.

Modality: the minimum number of instances of one entity that can be associated with an instance of another entity.

Modularity: a design principle that involves dividing a system into distinct and manageable sections or modules, each with its own specific responsibilities, which can be developed, tested and maintained independently, but function cohesively when combined.

Monopolize resources: the control or domination of the use of system resources (e.g. CPU, memory or network bandwidth) by a single process or user, often to the detriment of other processes or users, leading to inefficiency or system slowdowns.

Multi-core architectures: systems with multiple CPU cores on a single chip, allowing parallel execution of instructions and tasks.

Mutable: a set whose state or content can be changed after it has been created, allowing for modifications, e.g. adding, removing or altering elements within the object.

Mutator: a public method that allows external code to update or mutate the value of a private instance variable within an object; also known as "setter method" as it "sets" the value.

Network address translation: modifies the IP addresses of data packets as they pass through a router or firewall; this helps improve security and manages the limited number of IP addresses available through IPv4 by allowing multiple devices to share a single global IP address.

Network segmentation: dividing a computer network into smaller, distinct subnetworks to improve performance, security and management.

Network switch: a device that connects multiple other devices within a single segment of a computer network, only forwarding data to the specific device it is intended for.

Neural network: a computer algorithm that imitates the design of the human brain by using a set of interconnected nodes for the processing and analysing of data.

Nibble: 4 bits.

Node: a basic unit of a data structure, e.g. a linked list or tree, which contains data and typically links to or references other nodes.

Noise: unwanted electrical disturbances that can affect the integrity of signals being processed by a computer; this noise is not related to sound, but to variations in voltage or current that can disrupt the accurate transmission and processing of digital data.

Normalization: the process of organizing data in a relational database in a way to reduce data redundancy and to improve data integrity.

NoSQL database: a database designed to handle large volumes of data and diverse data types, structured differently from relational databases.

O(1) time complexity: describes an algorithm that takes the same amount of time to execute regardless of the size of the input data set.

Object-oriented programming: a form of programming that involves creating code for classes of objects, allowing many such objects to be created from a single code base, achieving a more modular and extensible software development process. It is like the idea of producing architectural blueprints, from which many similar houses can be constructed.

Observer pattern: provides a one-to-many link between objects to notify objects of changes in state via a subscription-style service.

Online analytical processing: the software technology you can use to analyse business data from different points of view.

Open addressing: a collision resolution method in hash tables where, instead of using structures like linked lists to store multiple items at the same index, any colliding item is placed into the next available open slot in the hash table itself, according to a probing sequence.

Operand: a value used in a mathematical expression.

Operator: a character that represents a mathematical, arithmetic or logical operation.

Outlier: a data point that deviates from the typical pattern of values in a data set, indicating a possible unusual or erroneous value that should be discounted.

Overriding: the process of providing a different implementation of a method in a subclass, which replaces the original implementation inherited from the superclass.

Packet switching: a method of sending data in small blocks, known as "packets", across a network. Each packet can take a different path to reach its destination.

Parallel arrays: a group of arrays of the same size, where the element at a given index in one of the arrays corresponds to another element at the same index in another array, like descriptions of a single entity.

Parallel processing: the ability of the GPU to perform many calculations simultaneously due to its highly parallel structure.

Parent: a node that has one or more nodes directly beneath it, connected by edges, and it directly controls these subsequent child nodes.

Partial functional dependency: when dependent attributes are partially determined by the determinant attributes.

Pattern recognition: identifying similarities in the details of problems.

Perceptron: the data structure at the heart of an artificial neural network; it represents a single artificial neuron that takes in multiple inputs and weights, and generates an output value.

Personal area network: a network for personal devices within the range of an individual person, usually connected with Bluetooth.

Physical schema: an implementation of logical schema into a specific DBMS (database management system), showing how data is stored, indexed or accessed.

Pixel: short for "picture element"; the smallest unit of a digital image or display, representing a single point in the image with a specific colour and intensity.

Plug and Play (PnP): a technology that allows the operating system to detect, configure and install drivers automatically for new hardware devices when they are connected to the computer, enabling them to work without requiring manual set-up by the user.

Pointer: a variable that stores the memory address of another variable, typically used in programming to reference, or access, the location of data stored in memory.

Polymorphism: meaning "many forms", it allows objects to exhibit different behaviours based on their specific class implementation while still adhering to a shared interface or contract.

Pop: a method for deleting the element from the top of a stack.

Primary key: a field that uniquely identifies a record in a table.

Problem specification: a short, clear explanation of an issue, which may include: a problem statement; constraints and limitations; objectives and goals; input and output specifications; and evaluation criteria.

Problem statement: a description of the problem itself, identification of who the solution is designed for, the issues encountered and what needs to be solved.

Procedure: a set of statements that can be grouped together and called in a program as needed; they don't return a value.

Product: the completed software only (in the internal assessment).

Proof of work: a consensus mechanism requiring cryptominers to solve complex problems to add a new block to the blockchain.

Protocol: a set of rules and standards that define how data is transmitted and received across a network for a given application.

Push: a method for inserting an element at the top of a stack.

Queue: an abstract data structure that works on the FIFO principle.

Quicksort: a sorting algorithm that repeatedly selects an element as a pivot and partitions the other elements into two sub-arrays (lists): one that includes elements that are smaller than the pivot and the other one that includes elements that are larger than the pivot.

Quotient: the result obtained when one number is divided by another, e.g. in the division of 15 by 3, the quotient is 5.

RAID (Redundant Array of Independent Disks): a data storage technology that combines multiple physical drives into a single logical unit to improve performance, provide redundancy and ensure data protection.

Record: one instance of an entity; a row in a table.

Recursion: a process that uses a function or procedure that is defined in terms of itself and calls itself.

Regression: machine learning where the output generated should be a numerical value.

Rehashing: a process in hash tables where the data is redistributed into a new, larger array to reduce the load factor and minimize collisions, maintaining efficient performance.

Reinforcement learning: machine learning by trial and error. Based on what it has learned at any moment in time, the algorithm selects an action to take in a given environment. The environment provides feedback (called a "reward"), which the algorithm will use to learn from and refine its decision-making process moving forward.

Relational operator: an operator used to compare values or expressions.

Relationship: a relation established between different tables, where the foreign key in one table refers to the primary key in another table.

Relative path: the location of a file relative to the current folder.

Rendering: the process of generating an image from a model by means of computer programs.

Root: the topmost node from which all other nodes descend, serving as the starting point for any traversal or operation within a binary search tree.

Router: a device that forwards data packets between computer networks, routing the traffic along the most efficient path.

Routing: the process of selecting paths along a computer network to send network traffic, based on the routing table, network performance and protocols.

R-squared value (or coefficient of determination): a statistical measure that indicates how well the linear regression model fits the data points given.

Runtime error: an error that occurs when executing a program; the program might stop unexpectedly.

Sampling: the process of converting a continuous analogue signal into a series of discrete digital values by measuring the signal's amplitude at regular intervals.

Second normal form: status of a relational database in which entities are in 1NF and any non-key attributes depend upon the primary key.

Security tokens: physical or digital devices that generate or store authentication credentials, such as one-time passwords or cryptographic keys, used to verify a user's identity and secure access to systems, networks or online services.

Selection: a conditional statement or decision statement, e.g. IF, CASE statements.

Selection sort: a sorting algorithm that repeatedly selects the smallest or largest element (ascending or descending order) from the unsorted part of the data structure and moves it to the sorted part.

Sequence: to execute instructions one after another in the given order.

Sequential access: a method of access where elements are checked one after another, from the beginning to the end of the data structure.

Server: a computer or device on a network that manages and provides various network resources on behalf of other computers (clients) on the network.

Set difference: the difference between two sets is a new set containing elements that are in the first set but not in the second set, effectively subtracting the elements of the second set from the first.

Set intersection: the intersection of two sets is a new set containing only the elements that are present in both of the original sets, identifying their common elements.

Set subset: a set where all elements of this set are also elements of another set, indicating that the first set is entirely contained within the second set.

Set union: the union of two sets is a new set containing all the elements that are in either of the original sets, effectively combining them without any duplicate elements.

Shaders and textures: techniques used in 3D rendering to apply effects, lighting and details to models.

Shift cipher: a type of substitution cipher, where each letter in the plaintext is shifted a certain number of positions down or up the alphabet.

Singleton pattern: a class that is designed only ever to have one instance instantiated throughout the lifecycle of the program.

Solution: the documentation and video submitted by the student for the internal assessment.

Spatial database: a database optimized to store and query data related to objects in space, including points, lines and polygons.

Spooling: the process of queuing data or tasks in a buffer, typically for input / output devices such as printers, so that they can be processed sequentially and at their own pace, allowing the system to continue working on other tasks in the meantime.

Stack: an abstract data structure that works on the LIFO principle.

Stack pointer: a register used to store the memory address of the last added data in a stack, or sometimes the first available address in a stack.

Stakeholder: an individual or groups of people within or outside an organization who are affected or think they are affected by a software development project.

Static: methods and variables that belong to the class, not the individual objects. Only one copy is created that is shared with all instances in common.

Static data structure: a data structure with predefined fixed size and elements stored in contiguous memory locations.

Statistical redundancy: the repetition of information within a data set that does not contribute to its uniqueness.

Stereo: a method of sound reproduction that uses two or more audio channels to create the perception of sound coming from different directions, enhancing the sense of spatial depth and dimension.

String: a data type used to represent a sequence of characters, digits and / or symbols.

Structural testing: testing concerned with the internal workings of the application – based on the code structure and internal pathways. This type of testing requires an understanding of the codebase and is used to ensure that all aspects of the code are properly tested. You are testing that all conditional branches execute correctly, and all error-handling code triggers when needed and responds appropriately.

Subtree: any node, along with its descendants, functioning as a standalone binary search tree, with its node acting as the root.

Supervised learning: when a machine learning algorithm is provided a data set of pairs of items, where the pair comprises a value and what response the network should provide if it sees that value. By learning the answers to the values given, the network will make generalizations to be able to estimate the answer when given a previously unseen value.

Table: a structure of rows and columns for storing a group of similar data.

Tensor: a mathematical term for an array with three or more dimensions. A single number (no dimensions) is known as a "scalar". A one-dimensional array of numbers is known as a "vector". A two-dimensional array of numbers is known as a "matrix". Three or more dimensions is known as a "tensor".

Termination condition: a condition in a loop that interrupts or stops the repetition.

Third normal form: status of a relational database in which entities are in 2NF and all non-key attributes are independent.

Trace table: a technique used to test an algorithm, and to predict how it will be run and how values of variables will change.

Transfer learning: when a previously trained machine learning model is applied to a similar yet new situation, context or problem. The goal is to speed up the training process by using an already trained model, even if the problem is slightly different.

Transform: to aggregate and transform data into a consistent format suitable for analysis.

Transitive dependency: a type of functional dependency that occurs when a non-prime attribute is dependent on another non-prime attribute, rather than on the primary key.

Trojans: deceptive programs that appear legitimate but carry hidden malicious code, which can create backdoors, steal data or cause harm once executed by the user.

Tuple: one instance of an entity; a row in a table.

Unary operator: an operator that requires one single operand.

Unordered set: a collection of unique elements where the elements do not have a specific order or sequence and their arrangement can vary each time they are accessed.

Unsupervised learning: a method of machine learning where the data set does not include the "answers" or expected outputs for the data provided. The algorithm will attempt to discover the patterns on its own.

Unwinding: a process occurring when the base case is reached, and the values are returned to build a solution.

Validation: a process to ensure input data is sensible or reasonable.

Variable: a designated memory location that stores a value that can change during the execution of a program.

Variable scope: the lifetime of a variable within a program; it determines whether you can access and modify the variable within a specific block of code.

Verification: a process to ensure input data is accurately copied from one source to another.

Vertex and pixel data: data used by the GPU to render 3D objects and images.

View: a virtual table based on the result set of a SELECT query. They do not store data themselves but provide a way to present the data from one or more tables in a customized manner.

Virtual memory: a memory-management technique that allows a computer to use more memory than is physically available by temporarily transferring data from RAM to disk storage, enabling the execution of larger programs and multitasking.

Virtual private network: a secure connection that runs across the internet to provide private communication between your network and a remote server.

Viruses: malicious software programs that attach themselves to legitimate files or programs and spread to other files or systems, often causing damage or disruption.

Volatile: a type of memory or storage that loses its data when the power is turned off.

Wide area network: a system that connects computers and other devices across a large geographic area, usually connecting multiple LANs together.

Winding: process occurring when recursive calls are made until the base case is reached.

Worms: self-replicating malware that spreads across networks without needing to attach to other programs, exploiting vulnerabilities to infect multiple systems.

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